4. Develop a program in C for simulating a Printer Queue using Queue data structure with the following operations:
a. Add a print job to the Printer Queue.
b. Process a print job from the Printer Queue (delete).
c. Display the number of print jobs currently waiting in the Queue.
d. Demonstrate Queue overflow and underflow conditions.
e. Exit.

```
/* Printer Queue implementation using arrays */


#include <stdio.h>

#define QUEUE_SIZE 3


int i, front = 0, rear = -1, item;

int q[QUEUE_SIZE];


void insert()

{

   if (rear == QUEUE_SIZE - 1)

   {

      printf("Printer Queue Overflow\n");

      return;

   }


   printf("Enter the print job ID\n");

   scanf("%d", &item);


   q[++rear] = item;

}







void del()

{

   if (front > rear)
```

```
    {
        printf("Printer Queue Underflow\n");
        return;
    }

    printf("Print job processed = %d\n", q[front]);
    front++;
}
void display()
{
    if (front > rear)
    {
        printf("Printer Queue is Empty\n");
        return;
    }
    printf("Contents of Printer Queue are:\n");

    for (i = front; i <= rear; i++)
    {
        printf("%d\t", q[i]);
    }
    printf("\n");

    printf("Number of print jobs waiting = %d\n",
          rear - front + 1);
}



int main()
{
    int ch;
```

```
    for (;;)
    {
        printf("\n1. Add Print Job");
        printf("\n2. Process Print Job");
        printf("\n3. Display Printer Queue");
        printf("\n4. Exit");
        printf("\nEnter your choice\n");
        scanf("%d", &ch);
        switch (ch)
        {
            case 1:
                insert();
                break;

            case 2:
                del();
                break;
            case 3:
                display();
                break;
            case 4:
                return 0;

            default:
                printf("Invalid choice\n");
        }
    }
}
```